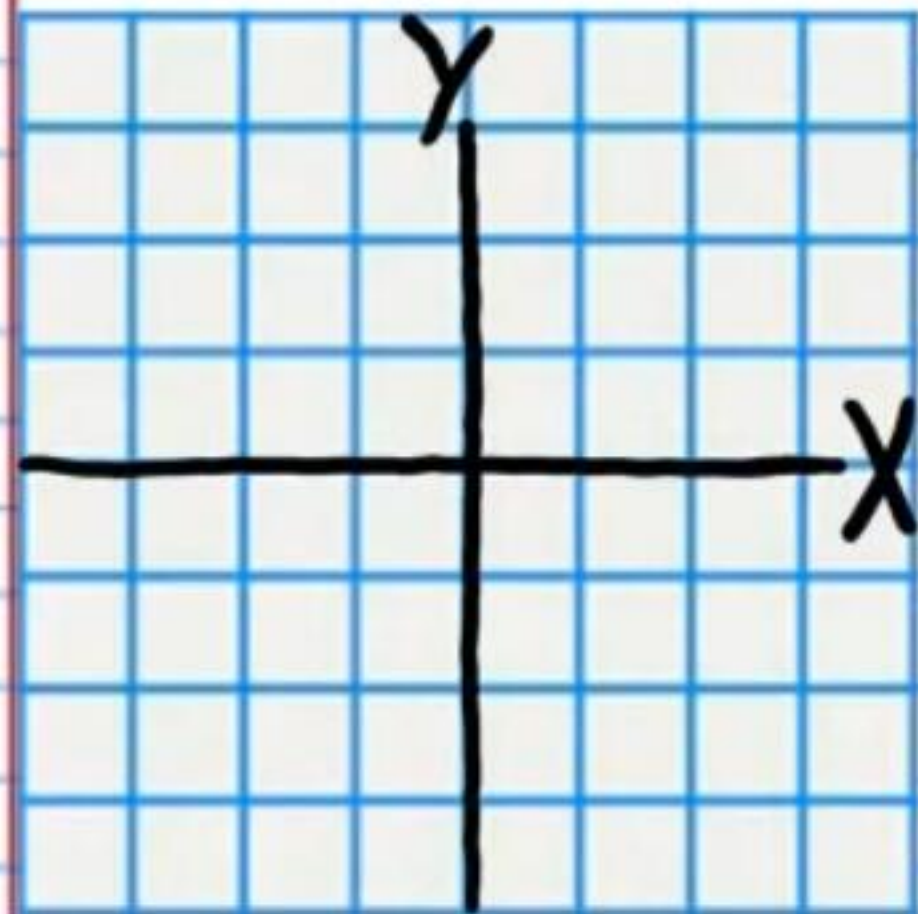


Final Test (Part II)

- I) This exam covers the second half of the course (content on Test III & Test IV).
 II) Write your name, the date, and the course title (i.e. Precalculus).
 III) Show all work.
 IV) Draw a rectangle around your final answer for each problem.
 V) A graphing calculator is required.
 VI) Each problem is worth four points for a total of 100 points.

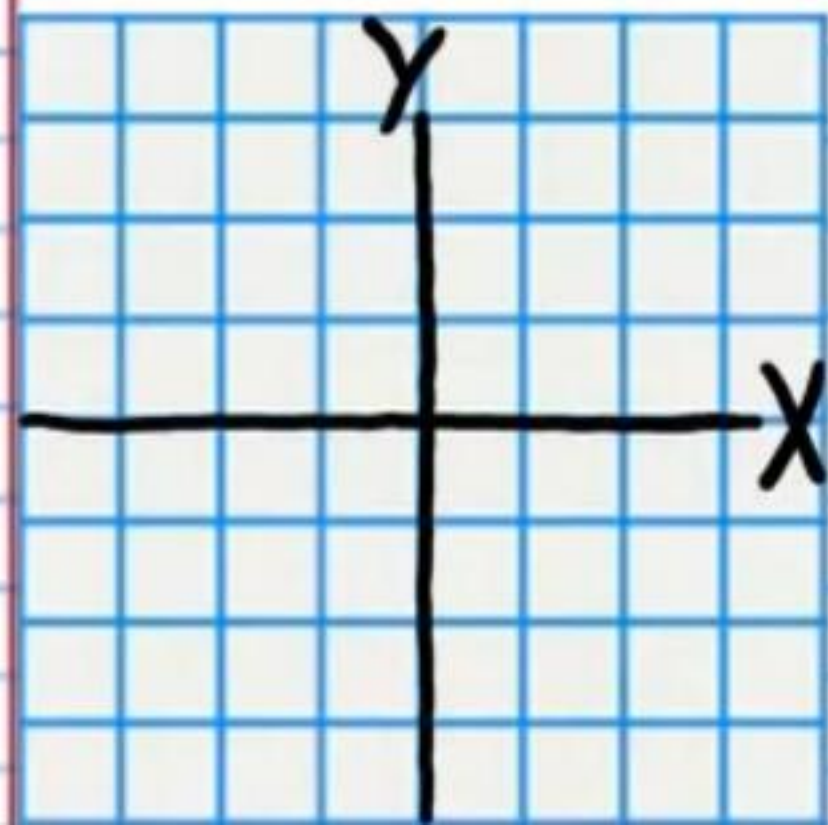
① Graph the following equation.

$$\cos \theta = \frac{1}{r}$$



② Graph the following complex number.

$$3 - 3i$$



③ Perform the following reflections.

i)

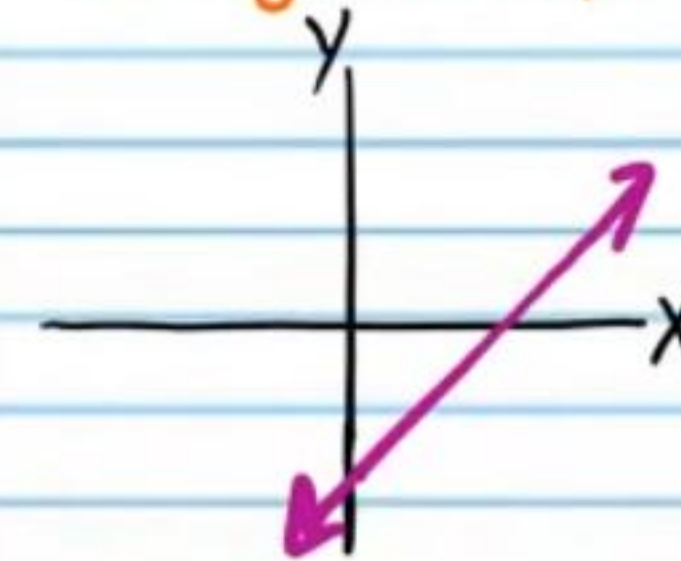
X	1	-5	4	8	2
Y	2	6	-1	-7	-2

Reflect across the origin

X					
Y					

ii) $\frac{1}{y} = \ln x$ Reflect across the x-axis

iii) Original Graph



Reflect across the y-axis



iv) $\cot x = \sin x$ Reflect across the origin

④ Create parametric equations for x and y using t as the parameter.

$$y = 4 - |x + 3|$$

⑤ Relations are written below in various forms. Write each inverse relation in its corresponding form.

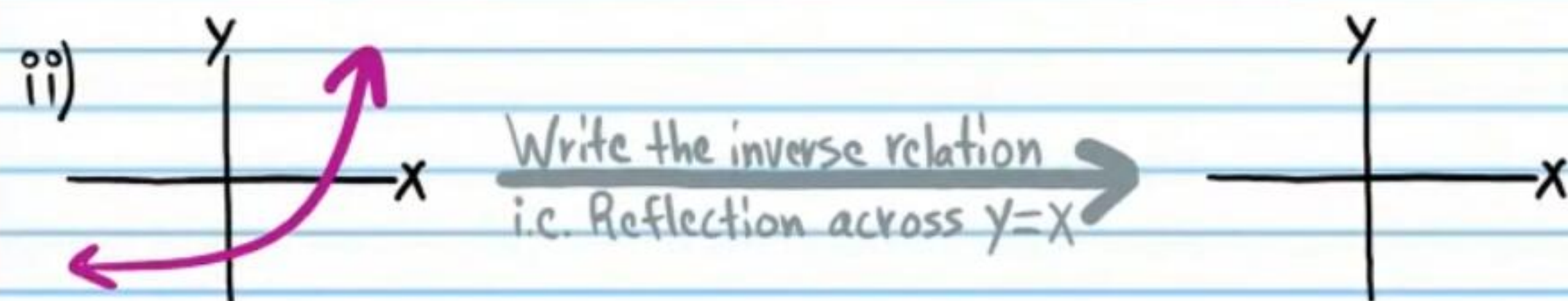
i)

x	-7	21	15	-11	0.1
y	13	-6	19	2	-3

 Write the inverse relation $\rightarrow$

x					
y					

i.e. Reflection across $y=x$



iii) $\frac{y}{x} - \sin x = 3$ Write the inverse relation $\rightarrow$
i.e. Reflection across $y=x$

iv) $\sqrt[3]{x} = 2xy + 10$ Write the inverse relation $\rightarrow$
i.e. Reflection across $y=x$

⑥ Determine if the function below is even, odd, or neither.

$$f(x) = x \sin x^2$$

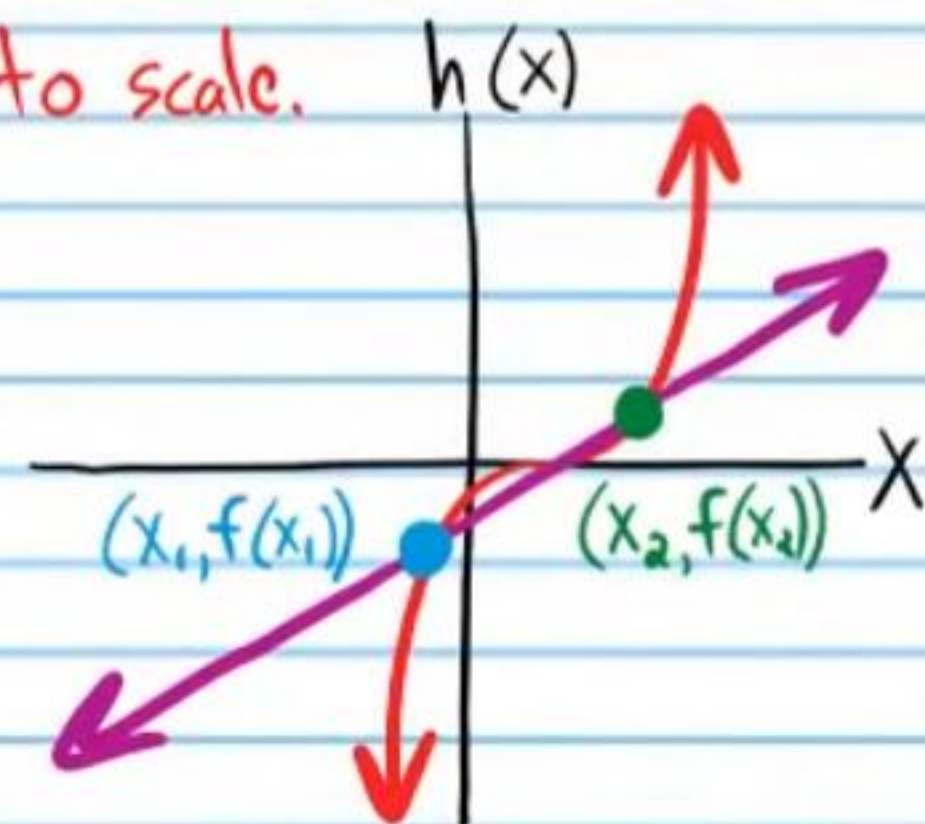
⑦ For the function below, find an algebraic expression representing the difference quotient (use "h" for "change in x"). It is not necessary to simplify the expression.

$$f(x) = x^2 + x$$

⑧ If $g(x) = x^2 + 4x - 1$ & $h(x) = 2x^2 + 11x + 11$, for what value(s) of x does $g(x) = h(x)$?

⑨ If $h(x) = (x-2)^3$, what is the average rate of change of the function per unit x from $x = -1$ to $x = 2$?

★ Not to scale.



⑩ Convert the equation below to polar form.

$$\left(\frac{y}{x}\right)^2 - 2 = x^2 + y^2 + x$$

⑪ If $f(x) = \frac{\sqrt{x-1}}{5} - 6$ and $f(b) = -4$, find b .

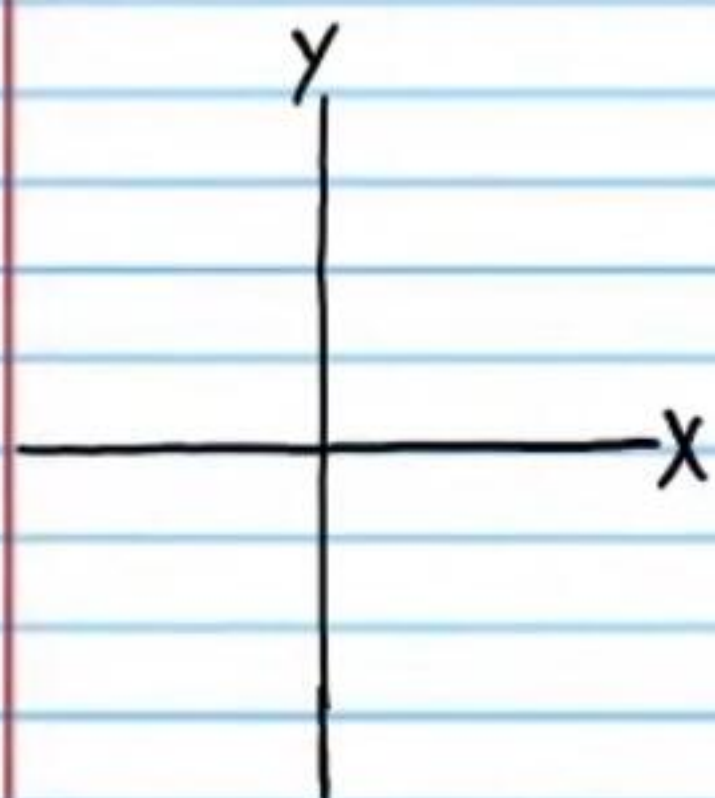
⑫ Convert the rectangular coordinates below to polar coordinates.
Use degree units for angle measures. Round to the nearest tenth if necessary.

$(-7, -10)$

Find r .

Find θ .

Graph

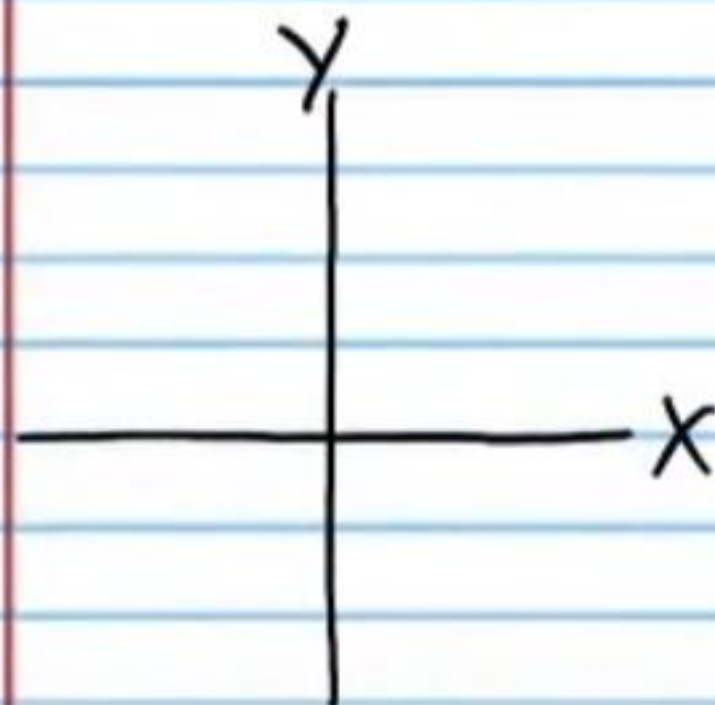


⑬ Convert the following polar coordinates to rectangular coordinates without a calculator.

$(6, 330^\circ)$

Find x . Find y .

Graph



For problem #'s 14-15, find an equation of the inverse function.

⑭ $f(x) = \ln(x+3)$

⑮ $f(x) = 10^{\frac{x}{2}+1}$

⑩ Show algebraically that "f" and "g" are inverse functions.

$$f(x) = \frac{x^5}{2} + 1 \quad \& \quad g(x) = \sqrt[5]{2x-2}$$

⑪ Find the intensity of sound required to produce the following decibels.

95 dB

⑮ The half life of copper-67 is 62 hours. If a physicist obtains 17 grams, how much will be left after 48 hours?

⑯ The population of a suspension of yeast after t hours can be predicted with the equation below.

$$P(t) = \frac{10,000}{1 + 24e^{-0.2t}}$$

i) What is the carrying capacity of the population?

ii) What is the initial population?

iii) What will the population be after 8 hours?

iv) Use your calculator's base "e" logarithm button (LN) and Logarithm Property IX to find the value of the logarithm below. Round to the nearest thousandth.

$$\log_6 927$$

20) i) In the following problem, "f" and "f⁻¹" are inverse functions.
If $f^{-1}(-100) = 2.87$, find $f(2.87)$.

ii) If "f" is an even function, and $f(-\frac{1}{5}) = -\frac{3}{7}$, find $f(\frac{1}{5})$.

iii) The table below has points collected from a graph that is symmetric about the origin. Fill the empty spaces with other points from the corresponding graph.

X	100	77	-10	-5	2														
Y	200	87	24	-11	-1														

iv) The table below has points collected from a graph of an odd function. Fill the empty spaces with other points from the graph.

X	-7	-9	-12	81	13														
f(x)	-3	-11	37	14	-6														

21) Use the following equation to answer the questions below.

$$3x^3 = y$$

i) Is the graph of the equation symmetric about the y-axis? *Demonstrate algebraically.*

ii) Is the graph of the equation symmetric about the origin? *Demonstrate algebraically.*

22) i) If "f" is an odd function, and $f(800) = -300$, find $f(-800)$.

ii) Relations are written below in table form. Some of the tables could represent functions that are one-to-one, some of them could represent functions that are not one-to-one, and some cannot represent functions. For each table, write "1-to-1 function," "function," or "not a function."

X	23	23	16	17	20
Y	18	-10	17	-16	29

X	16	15	14	13	16
Y	82	81	80	79	78

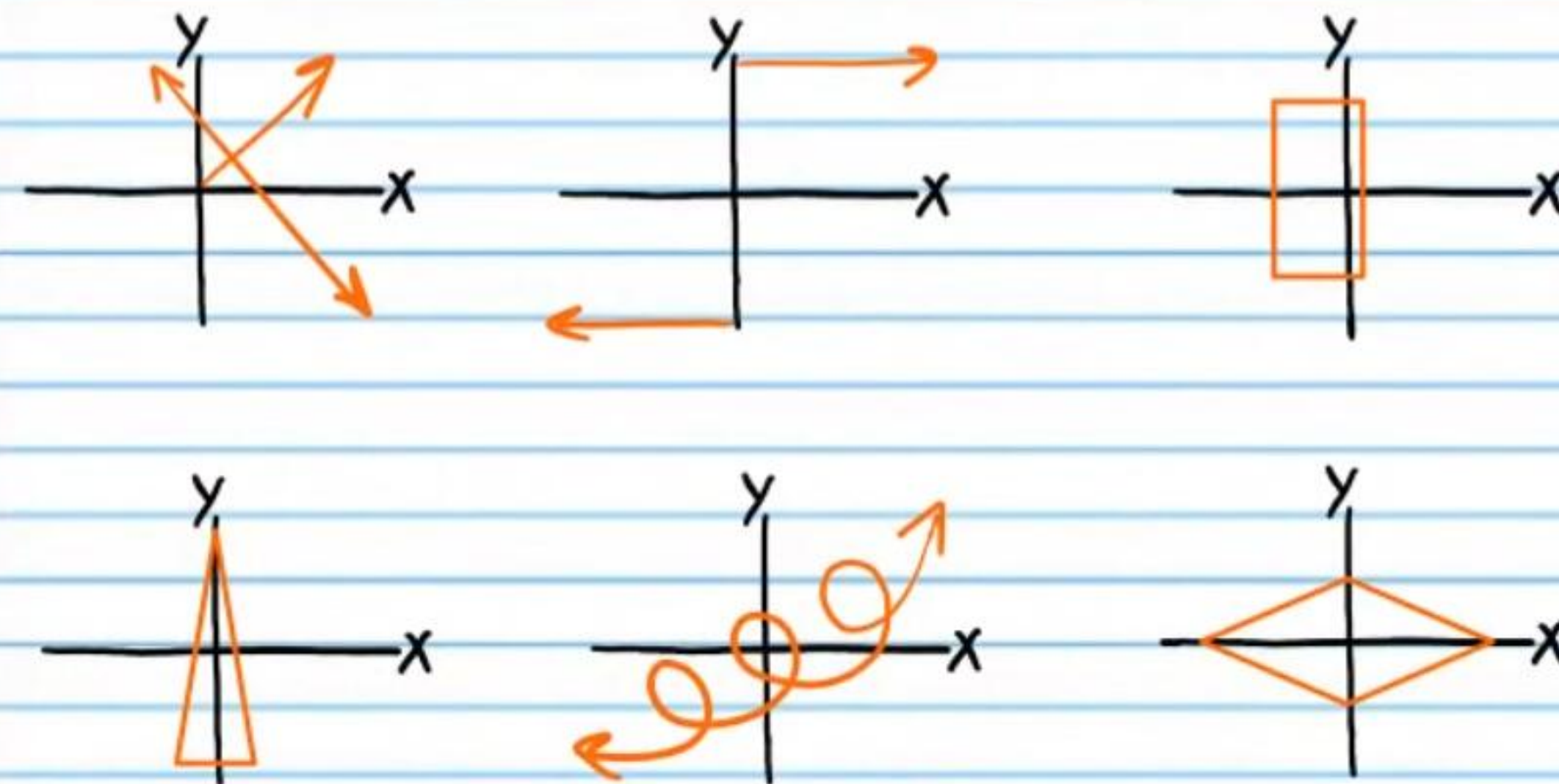
X	$\frac{1}{2}$	13	14	21	-1
Y	$\frac{2}{3}$	8	10	28	-3

X	800	523	117	100	32
Y	700	231	420	420	420

iii) Determine if the graphs below are even functions, odd functions, or neither.

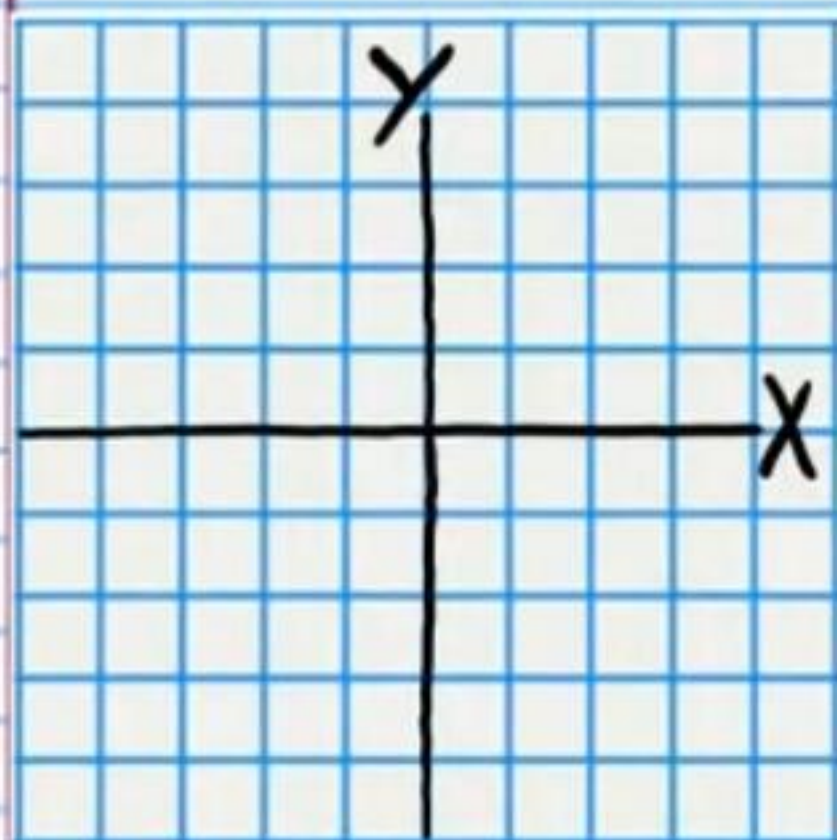


iv) Relations are written below in the form of graphs. If a relation appears to be symmetric about the x-axis, write "x." If a relation appears to be symmetric about the y-axis, write "y." If a relation appears to be symmetric about the origin, write "O." If the relation shows no apparent symmetry, write "N." *Some graphs might display multiple types of symmetry.*



②③ Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot. Then eliminate the parameter.

$$X = t + 1 \quad Y = \frac{1}{t + 2} \quad \text{Point \#1} \quad \text{Point \#2}$$



Point #3

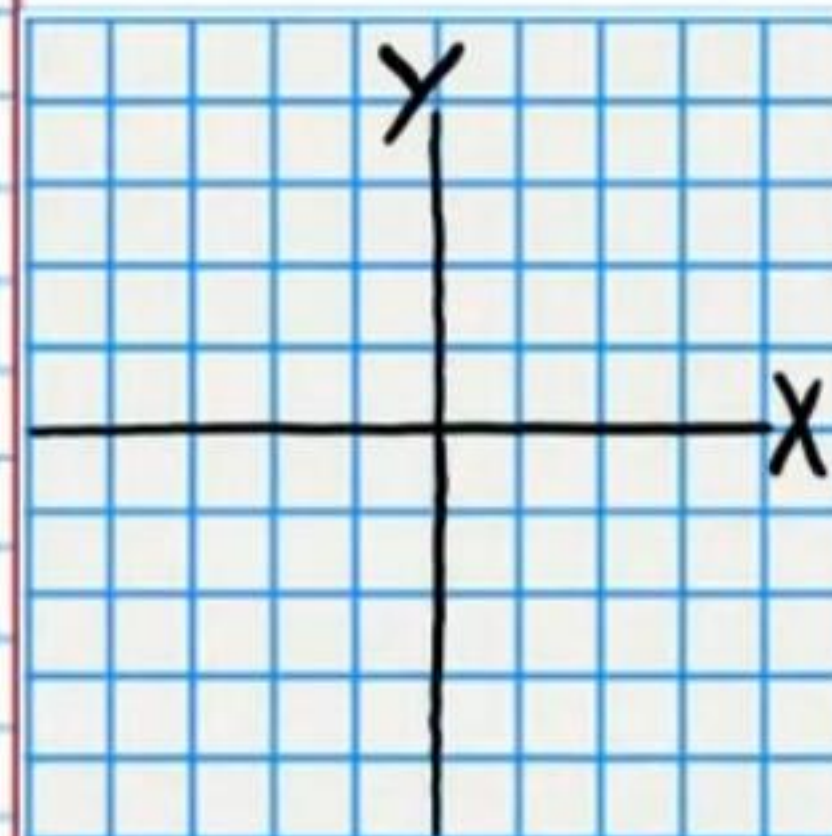
Point #4

Eliminate the parameter

$$X = t + 1 \quad Y = \frac{1}{t + 2}$$

②④ Graph the following equation using radian units. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

$$r = \frac{8}{4 + 3\sin\theta} \quad \text{1st Point} \quad \text{2nd Point}$$

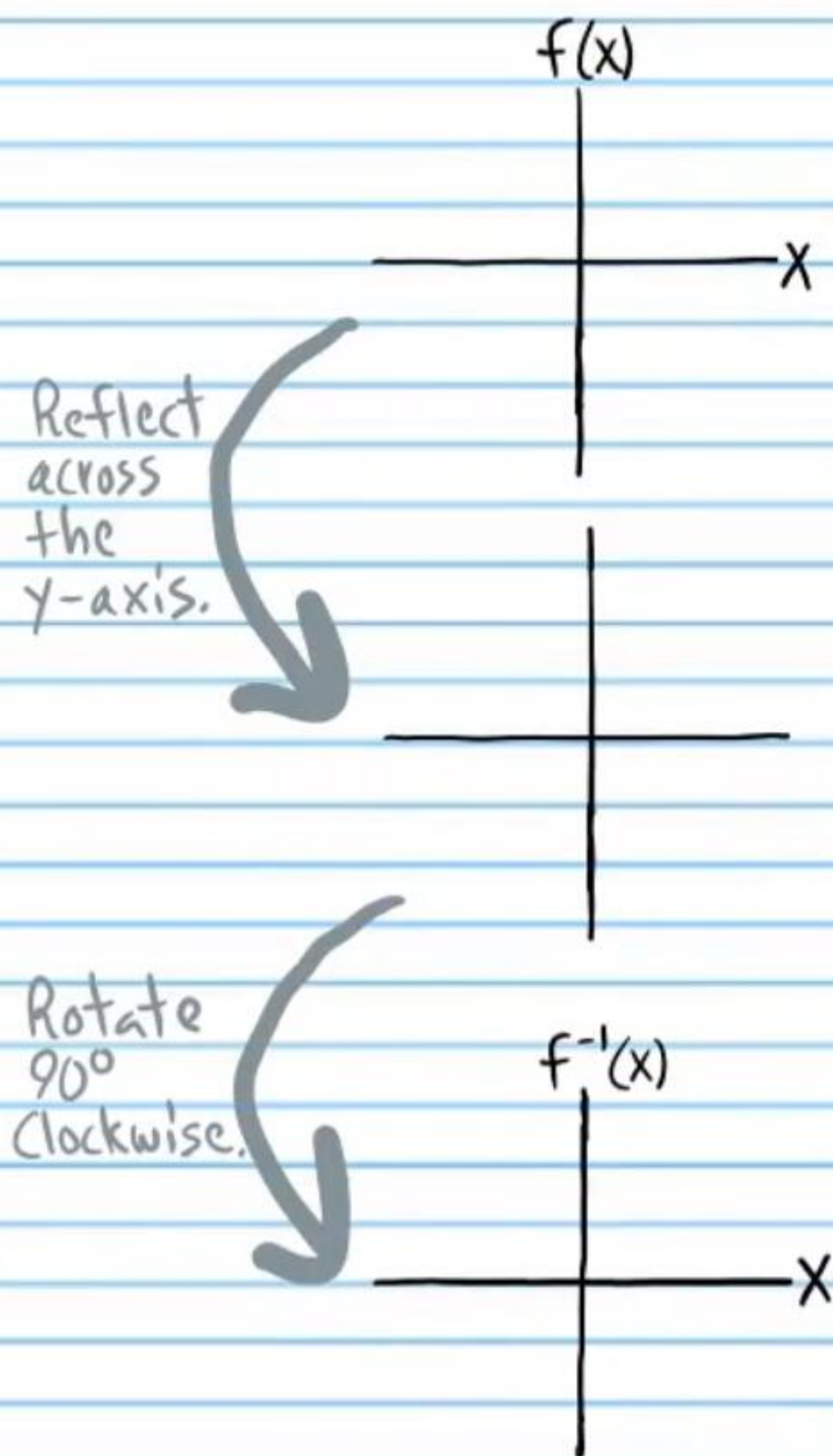


3rd Point

4th Point

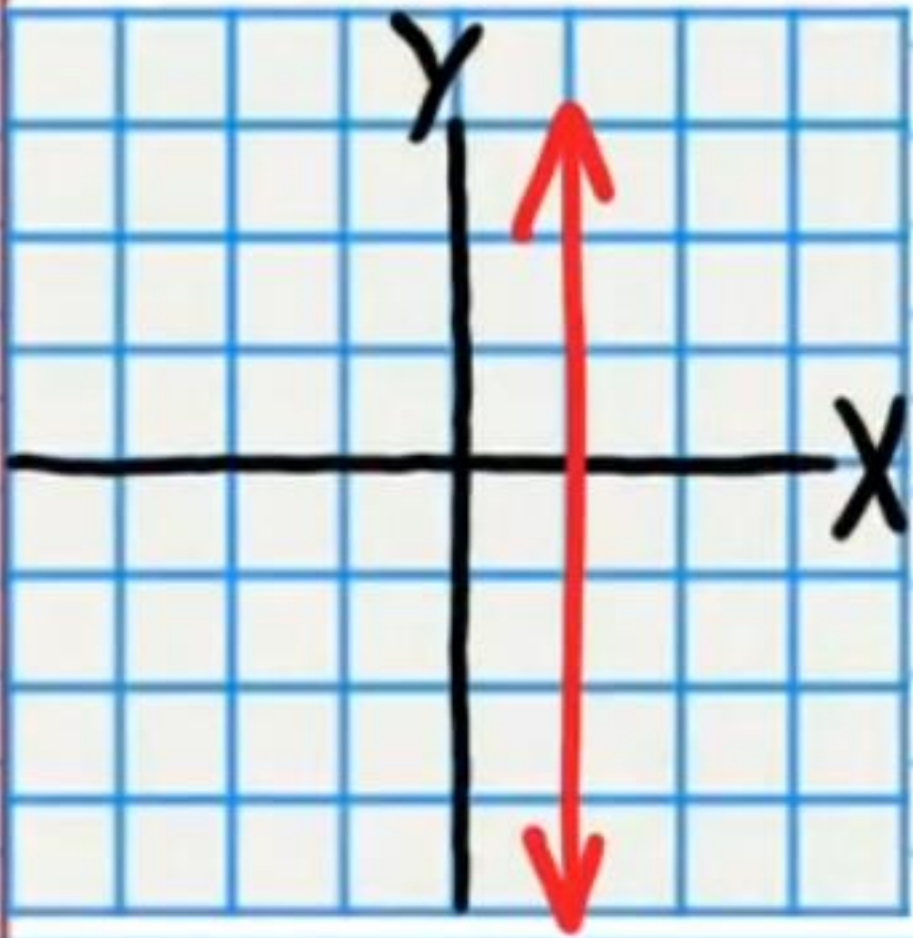
Q5 Find an equation of the inverse function. Restrictions must be used on the inverse function so that it is perfectly matched with the original function. Circle or highlight the parts of the inverse graph that must be omitted and use inequality notation next to the inverse equation to indicate restrictions.

$$f(x) = \sqrt{x} - 1$$

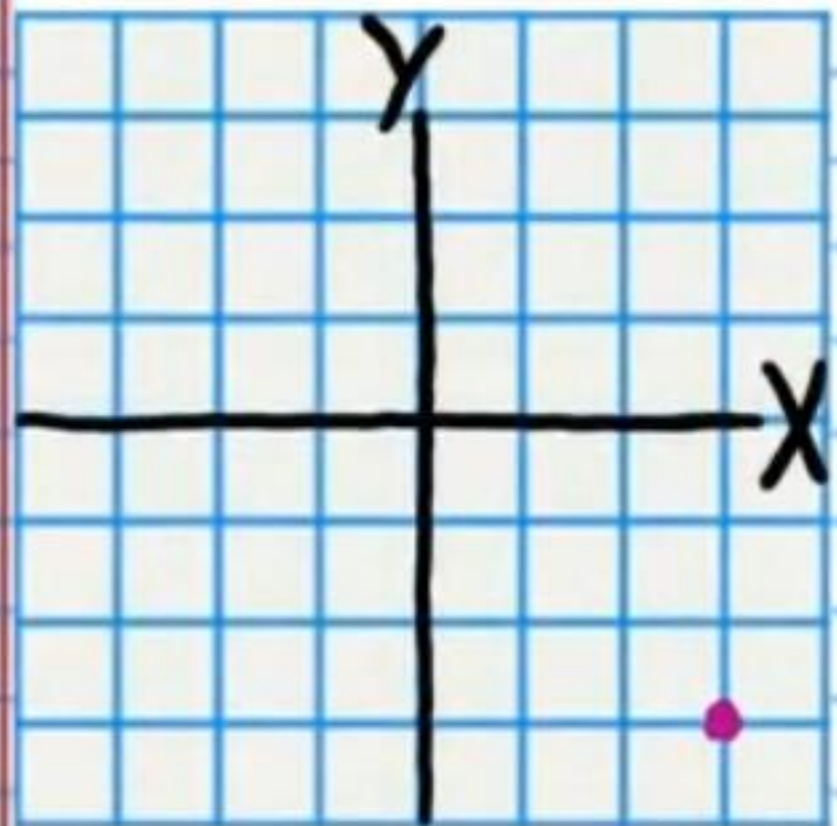


Answers

①



②

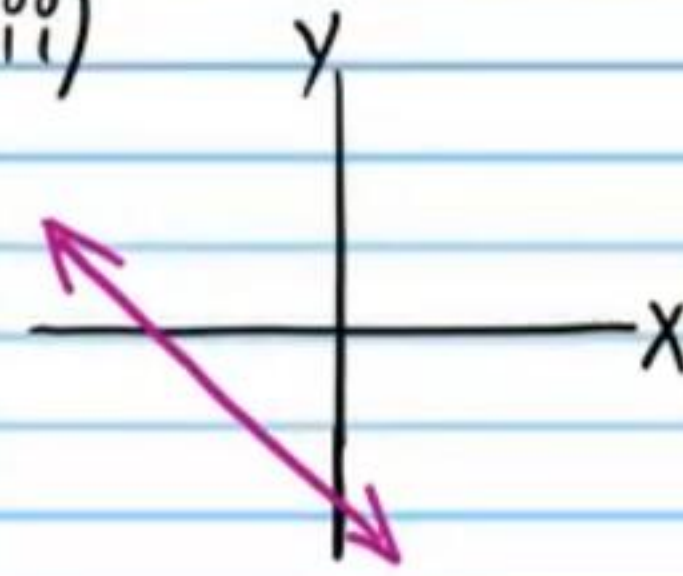


③ i)

x	-1	5	-4	-8	-2
y	-2	-6	1	7	2

ii) $-\frac{1}{y} = \ln x$

iii)



iv) $\cot(-x) = \sin(-x)$

$-\cot x = -\sin x$

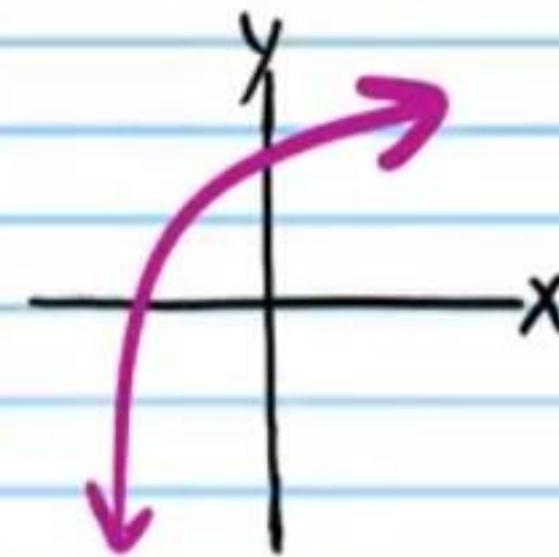
$\cot x = \sin x$

④ $x = t, y = 4 - |t + 3|$

⑤ i)

x	13	-6	19	2	-3
y	-7	21	15	$-\pi$	0.1

ii)



iii) $\frac{x}{y} - \sin y = 3$

iv) $\sqrt[3]{y} = 2yx + 10$

$$\textcircled{6} f(-x) = (-x)\sin(-x)^2$$

$$f(-x) = -x\sin x^2$$

$$f(-x) = -f(x)$$

Odd

$$\textcircled{7} DQ = \frac{(x+h)^2 + x+h - (x^2+x)}{h}$$

$$\textcircled{8} x = -3, x = -4$$

$$\textcircled{9} 9$$

$$\textcircled{10} \tan^2 \theta - 2 = r^2 + r \cos \theta \quad \text{or} \quad \left(\frac{\sin \theta}{\cos \theta}\right)^2 - 2 = r^2 + r \cos \theta$$

$$\textcircled{11} b = 101$$

$$\textcircled{12} (-\sqrt{149}, 55^\circ) \text{ or } (\sqrt{149}, 235^\circ) \text{ or } (\sqrt{149}, -125^\circ) \text{ * Other variations are possible.}$$

$$\textcircled{13} (3\sqrt{3}, -3)$$

$$\textcircled{14} f^{-1}(x) = e^x - 3$$

$$\textcircled{15} f^{-1}(x) = 2(\log x - 1) \quad \text{or} \quad f^{-1}(x) = 2\log x - 2$$

$$\textcircled{16} f(g(x)) = f(\sqrt[5]{2x-2}) = \frac{(\sqrt[5]{2x-2})^5}{2} + 1 = \frac{2x-2}{2} + 1$$

$$= x - 1 + 1$$

$$= x \checkmark$$

$$g(f(x)) = g\left(\frac{x^5}{2} + 1\right) = \sqrt[5]{2\left(\frac{x^5}{2} + 1\right) - 2} = \sqrt[5]{x^5 + 2 - 2}$$

$$= \sqrt[5]{x^5}$$

$$= x \checkmark$$

Yes

$$\textcircled{17} 10^{-2.5} \text{ W/m}^2$$

$$\textcircled{18} 9.94 \text{ grams}$$

$$\textcircled{19} \text{ i) } 10,000 \text{ organisms}$$

$$\text{ii) } 400 \text{ organisms}$$

$$\text{iii) } 1,711 \text{ organisms}$$

iv) $\boxed{3.813}$

20 i) $f(2.87) = \boxed{-100}$

ii) $f(\frac{1}{5}) = \boxed{-\frac{3}{7}}$

iii)

X	100	77	-10	-5	2	-100	-77	10	5	-2
Y	200	87	24	-11	-1	-200	-87	-24	11	1

iv)

X	-7	-9	-12	81	13	7	9	12	-81	-13
f(x)	-3	-11	37	14	-6	3	11	-37	-14	6

21 i) $y = 3(-x)^3$ ii) $-y = 3(-x)^3$

$y = 3(-x^3)$

$-y = 3(-x^3)$

$y = -3x^3$

$-y = -3x^3$

$\boxed{\text{No}}$

$y = 3x^3$

$\boxed{\text{Yes}}$

22 i) $f(-800) = \boxed{300}$

ii) Not a Function Not a Function

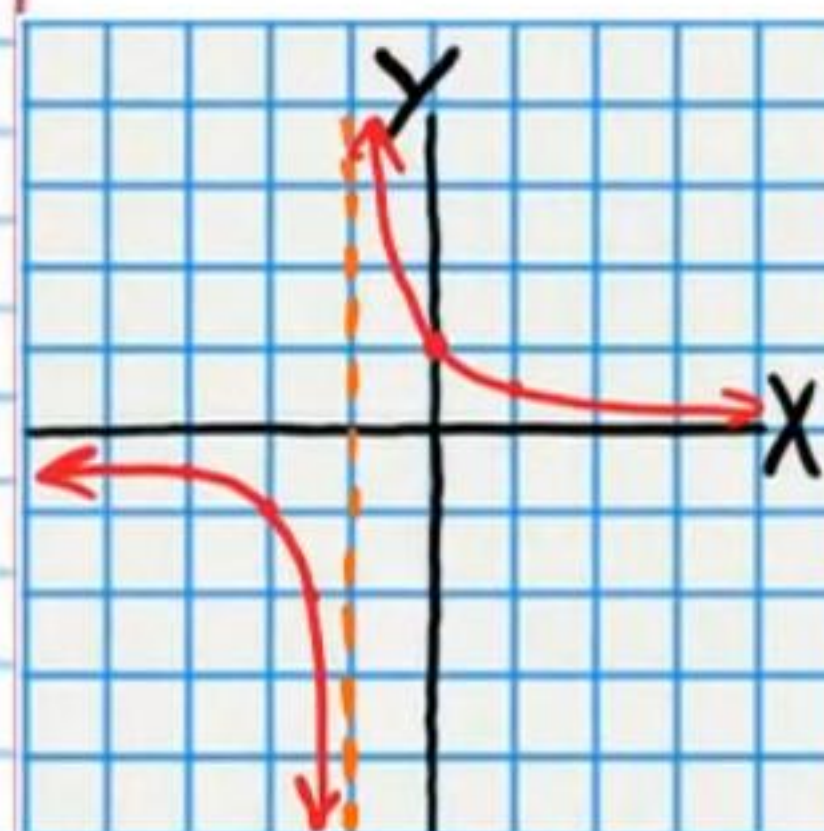
1-to-1 Function Function

iii) Odd Neither Even Neither

iv) N O X

Y N XYO

23

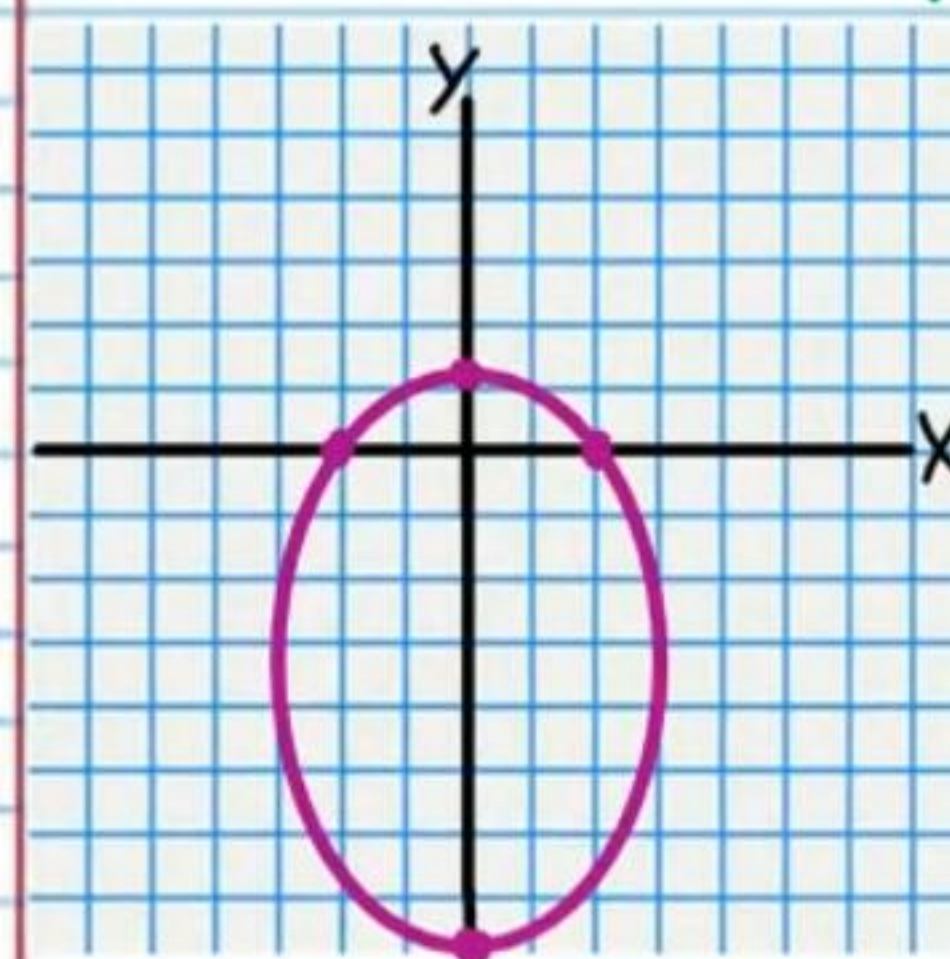


Other points are possible. Only 4 are necessary.

$\boxed{(1, \frac{1}{2}) (0, 1) (2, \frac{1}{3}) (-2, -1) (-3, -\frac{1}{2})}$

$\boxed{y = \frac{1}{x+1}}$

24



Other points are possible. Only 4 are necessary.

$\boxed{(\frac{8}{7}, \frac{\pi}{2}) (2, 0) (8, \frac{\pi}{2}) (2, \pi)}$

$\boxed{(\frac{16}{11}, \frac{\pi}{6}) (3\frac{1}{5}, -\frac{\pi}{6})}$

25 $\boxed{f^{-1}(x) = (x+1)^2, x \geq -1}$

